

VIVEKANANDA GLOBAL UNIVERSITY, JAIPUR

(Established by Act 11/2012 of Rajasthan Govt. Covered u/s 2 (f) of UGC Act, 1956)

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Unit-1

Definition of Algorithm:

Algorithm is a step-by-step procedure, which defines a set of instructions to be executed in a certain order to get the desired output. Algorithms are generally created independent of underlying languages, i.e. an algorithm can be implemented in more than one programming language.

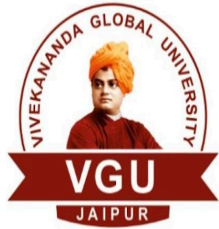
From the data structure point of view, following are some important categories of algorithms –

- Search – Algorithm to search an item in a data structure.
- Sort – Algorithm to sort items in a certain order.
- Insert – Algorithm to insert item in a data structure.
- Update – Algorithm to update an existing item in a data structure.
- Delete – Algorithm to delete an existing item from a data structure.

Characteristics of an Algorithm:

Not all procedures can be called an algorithm. An algorithm should have the following characteristics –

- **Unambiguous** – Algorithm should be clear and unambiguous. Each of its steps (or phases), and their inputs/outputs should be clear and must lead to only one meaning.
- **Input** – An algorithm should have 0 or more well-defined inputs.
- **Output** – An algorithm should have 1 or more well-defined outputs, and should match the desired output.
- **Finiteness** – Algorithms must terminate after a finite number of steps.
- **Feasibility** – Should be feasible with the available resources.
- **Independent** – An algorithm should have step-by-step directions, which should be independent of any programming code.



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Examples:

Algorithm 1: Add two numbers entered by the user

Step 1: Start

Step 2: Declare variables num1, num2 and sum.

Step 3: Read values num1 and num2.

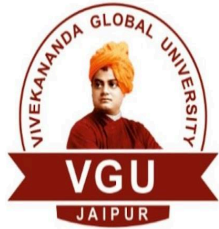
Step 4: Add num1 and num2 and assign the result to sum.

```
sum←  
num1  
+num
```

```
sum←num1+num2
```

Step 5: Display sum

Step 6: Stop



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Algorithm 2: Find Roots of a Quadratic Equation $ax^2 + bx + c = 0$

Step 1: Start

Step 2: Declare variables a, b, c, D, x1, x2, rp and ip;

Step 3: Calculate discriminant

$D \leftarrow$

$b^2 - 4ac$ Step 4: If D

≥ 0

$r1 \leftarrow (-b + \sqrt{D}) / 2a$

$r2 \leftarrow (-b - \sqrt{D}) / 2a$

Display r1 and r2 as roots.

Else

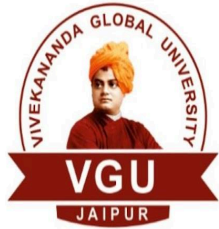
Calculate real part and imaginary part

$rp \leftarrow -b / 2a$

$ip \leftarrow \sqrt{-D} / 2a$

Display $rp + j(ip)$ and $rp - j(ip)$ as roots

Step 5: Stop



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Algorithm Analysis:

Efficiency of an algorithm can be analyzed at two different stages, before implementation and after implementation. They are the following –

A Priori Analysis – This is a theoretical analysis of an algorithm. Efficiency of an algorithm is measured by assuming that all other factors, for example, processor speed, are constant and have no effect on the implementation.

A Posterior Analysis – This is an empirical analysis of an algorithm. The selected algorithm is implemented using programming language. This is then executed on target computer machine. In this analysis, actual statistics like running time and space required are collected.

Pseudocode:

- A program planning tool where program logic is written in an ordinary natural language using a structure that resembles computer instructions
- “ Pseudo” means imitation or false and “ Code” refers to the instructions written in a programming language. Hence, pseudocode is an imitation of actual computer instructions
- Because it emphasizes the design of the program, pseudocode is also called Program Design Language (PDL)